Practice Quiz 09A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

Images and inverse images; the geometric sum formula.

1. (a) Let $f : \{1, 2, 3, 4\} \to \{a, b, c\}$ be given by

$$f(1) = a, \quad f(2) = b, \quad f(3) = b, \quad f(4) = c.$$

For $X = \{1, 3\}$, compute $f(X)$ and $f^{-1}(f(X))$. State whether $f^{-1}(f(X)) = X$.

Solution The image is $f(X) = \{f(1), f(3)\} = \{a, b\}$. The elements mapped into $\{a, b\}$ are $1, 2, 3$, so

$$f^{-1}(f(X)) = \{1, 2, 3\} \neq \{1, 3\} = X.$$

The extra element 2 appears because $f(2) = f(3) = b$. ∎

(b) State the geometric sum formula for $1 + r + \cdots + r^n$, including its hypotheses, and prove it by induction on n.

Solution For every real $r \neq 1$ and every integer $n \geq 0$,

$$\sum_{i=0}^{n} r^i = \frac{r^{n+1} - 1}{r - 1}.$$

Fix a real $r \neq 1$. For $n = 0$, the left side is 1 and the right side is $(r - 1)/(r - 1) = 1$.

Suppose the formula holds for an integer $n \geq 0$. Adding the next term gives

$$\begin{aligned} \sum_{i=0}^{n+1} r^i &= \frac{r^{n+1} - 1}{r - 1} + r^{n+1} \\ &= \frac{r^{n+1} - 1 + r^{n+1}(r - 1)}{r - 1} = \frac{r^{n+2} - 1}{r - 1}. \end{aligned}$$

This is the formula for $n + 1$. Induction proves it for every $n \geq 0$, and $r \neq 1$ was arbitrary. ∎

Solutions revised using GPT-6 Astra in the Codex harness.